

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461057.9499 +/- 0.0068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.044
Transit depth (Rp/Rs)^2	2.21 +/- 1.32 [%]
Orbital Inclination (inc)	88.37 +/- 2.78
Airmass coefficient 1 (a1)	0.119 +/- 0.036
Airmass coefficient 2 (a2)	1.8 +/- 0.62
Scatter in the residuals of the lightcurve fit is	30.2 %
Best Comparison Star	#2 - [538, 52]
Optimal Aperture	4.6
Optimal Annulus	9.97
Transit Duration (day)	0.0936 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.044 vs 0.1214 (deviation 0.54σ)
Duration fit vs published	0.0936 d vs 0.0742 d (deviation 1.81σ)
Mid-time O–C	18.4 min
Depth SNR	2.9
Residual scatter	30.2 %

Light curve

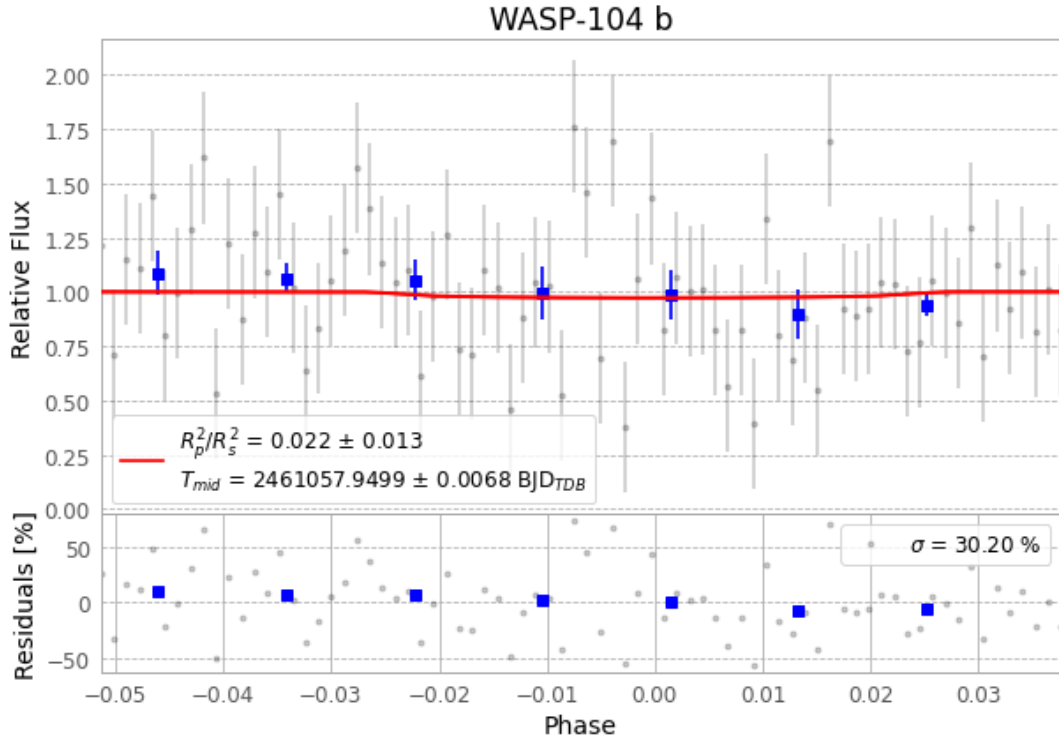


Figure 1 — FinalLightCurve_WASP-104 b_2026-01-17.png (SHA-256 95c301cb6c7f0da6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 282], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9ffa2cb30d7bd825...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

76 FITS frames (50 MB), WASP-104260117083015.FITS → WASP-104260117121516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-260117.log (28ea54a783db...), FinalLightCurve_WASP-104 b_2026-01-17.png (95c301cb6c7f...), FinalLightCurve_WASP-104 b_2026-01-17.csv (d653b1fe1aba...)

Compute resources

Wall time 176.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:09Z.